



Course name	Network Architectures and Protocols / Computer networks
The topic of the document	Laboratory exercise booklet for university students
Laboratory exercise booklet no.	8.
The title and the topic of the laboratory exercise	Transport layer mechanisms

1. Preparation and implementation requirements:

- The student is obliged to read in advance the content of the present laboratory exercise booklet, to do the preparations for the laboratory and to observe the rules defined for the laboratory.
- The students should read the description of the special terms, mechanisms listed in the „Theoretical background information” chapter to accomplish the laboratory tasks successfully.
- The appearance happens at the beginning of the laboratory, delay can be allowed only in justified cases.
- The use of students’ own mobile phone and PC is forbidden in the laboratory, except the case when the laboratory leader permits their use for the students because of the type of the laboratory exercise to be done.
- Browsers can be used exclusively for the access to information which is necessary for the accomplishment of the laboratory exercise, in the case of the laboratory leader’s permission or instruction.
- By the end of the class the laboratory leader evaluates each participant’s activity based on the composed questions which are included in this present material and further possible questions.
- The laboratory will be successful, if the student provides a correct answer to each question on the basis of the laboratory exercises which are carried out on the premises.
- The replacement of the laboratory is possible at dates of time determined by the laboratory leader.

2. Theoretical background information:

During the laboratory the following special terms will be used in practice:

- transport layer properties ([ppt slides of the lecture](#), slide no. 226-228.)
- datagram ([ppt slides of the lecture](#))
- segment ([ppt slides of the lecture](#))
- UDP segment structure ([ppt slides of the lecture](#), slide no. 230-231.)
- TCP segment structure ([ppt slides of the lecture](#), slide no. 232-234.)
- TCP flag bits/flags ([ppt slides of the lecture](#), slide no. 234.)
- port numbers ([ppt slides of the lecture](#), slide no. 243.)
- socket pair ([Wikipedia](#))
- list of well-known and registered services ([internet](#))



- Wireshark filters (capture filters, display filters)

During the laboratory the following mechanisms/technics will be used in practice:

- data segmentation (fragmentation) (ppt slides of the lecture, slide no. 150-151.)
- TCP three-way handshake (ppt slides of the lecture, slide no. 235.)
- TCP four-way handshake (ppt slides of the lecture, slide no. 236.)
- adatfolyam vezérlés/data flow control (ppt slides of the lecture, slide no. 228.)
- multiplexing/demultiplexing (ppt slides of the lecture, slide no. 228.)
- TCP sliding window (ppt slides of the lecture, slide no. 237.)
- TCP performance control (ppt slides of the lecture, slide no. 240-242.)
- TCP congestion control (ppt slides of the lecture)
- port scanning (Wikipedia)
- netstat (Wikipedia)
- netcat (Wikipedia)
- nmap (Wikipedia)
- NAT/PAT translation (ppt slides of the lecture, slide no. 181-182.)

3. The following tasks have to be performed on the site:

3.1.	List of well-known and registered services.
	• Display the content of <code>/etc/services</code> file, then search the default port number of some well-known services: FTP, Telnet, SSH, SMTP, POP3, IMAP, DNS, HTTP, HTTPS.
3.2.	Monitoring TCP connection establishment/connection termination (Wireshark analysis).
	• Start capturing on the Ethernet interface of your VM, then open the browser program of your VM and download the opening page of www.unideb.hu site. • Stop capturing, then analyse the captured packets. • Use Display Filter for the network traffic in order that Wireshark could present the communication between your own VM and the TCP port 80 of the destination host IP address 193.225.40.94. • Modify the Display Filter rule in order that Wireshark could list only TCP segments with SYN flag, then look at the same with FYN flag as well.
3.3.	Review of the Finite State Machine (FSM) which manages the TCP connection
	• Identify the TCP connection establishment phase or connection termination phase on the Finite State Machine managing the TCP connection.
3.4.	Display network connections.



	List your active network connections on your VM, then determine if there is an active web connection with the destination host IP address 193.225.40.94.
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3.5.	Creation of a chat program.
	Build a client and server-based chat environment with the help of <i>netcat</i> tool.

3.6.	Implementation of file transfer.
	- Create a text file, then copy it via network to other VM with the help of <i>netcat</i> tool. - Can we open the copied file?

3.7.	Port scanning with <i>netcat</i> tool on two machines.
	Open port 9999 on the server, then start <i>netcat</i>-based port scanning on the client.

3.8.	Port scanning with <i>nmap</i> tool.
	Perform port scanning on <i>freemail.hu</i> server with the help of <i>nmap</i> tool.

4. The following questions have to be answered by the end of the laboratory:

4.1.	A) On which port number(s) does FTP communicate?
	B) Which command can we use for filtering a given service name?

4.2.	A) How many Wireshark filter types do you know? Name them.
	B) How can we make a diagram-based statistical report with Wireshark regarding the captured packets?

4.3.	How many system states does the Finite State Machine (FSM) contain which manages TCP connection?
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	B) What is the difference between an active and a passive TCP connection termination?
4.4.	Can we use <i>netstat</i> command in Windows environment?
	B) How (using which command and what way) can we determine if we have an active web connection with destination host IP address 193.225.40.94?
4.5.	A) Can we use the <i>netcat</i> tool for implementing a chat program?
	B) How could we block the running of such a chat environment in the most simple way?
4.6.	A) Can file transfer happen above UDP?
	B) In the case of <i>Netcat</i> -based file transfer is server or client side necessary to implement first? Justify your answer.
4.7.	A) Which option can we use for setting the scanned server side in listening mode?
	B) Can we use the <i>netcat</i> tool for scanning several ports which are opened at the same time?
4.8.	A) Can we use <i>nmap</i> port for opening?
	B) Which port numbers are open on <i>freemail.hu</i> server?